An Adaptive Multi-Horizon Decision Transformer for Cross-Asset Portfolio Management

Market-Wide Representation Learning with Per-Asset

Temporal and Credit-Assignment Adaptivity

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Abstract

Portfolio management policies today largely assume that every asset in a universe deserves the same temporal receptive field and the same credit-assignment horizon. In practice, a broad-market ETF, a momentum stock reacting to a single earnings print, and a supply-chain-linked semiconductor name all demand different amounts of memory and different reward horizons, and their relevance to one another shifts with market regime. This proposal describes an Adaptive Multi-Horizon Decision Transformer (AMH-DT): a single pipeline that first builds a market-wide representation of all assets jointly, learns their dynamic cross-asset relationships through attention, and then adaptively routes each asset's current behaviour to a temporal-horizon specialist before fusing all specialist outputs into one portfolio decision. I position this design against the closest prior work, including EarnMore [1], STHformer [2], Action Q-Transformer [3], and Mixture-of-Experts trading systems such as MoEDRLPM [4], LLMoE [5], and DeepUnifiedMom [6], and identify the specific gap each leaves open. I also flag one design risk in the routing stage as originally sketched, and propose a modification that keeps the core idea intact while removing a likely source of training instability.

1 Motivation and Problem Statement

Two assumptions are baked into most sequence-model approaches to portfolio management, and both are wrong often enough to matter:

1. **Fixed temporal receptive field:** The same context window and attention span are applied to every asset, regardless of whether that asset's price behaviour is driven by slow macro trends or by a single fast-moving catalyst.

2. **Independent asset processing:** Many Decision-Transformer-style trading policies [7] process each asset's trajectory in isolation, so sector co-movement, supply-chain dependency, and macro-sensitivity are not represented inside the policy at all.

3. The mere assumption that the largely unsatisfying attempts to apply the two most natural and well performing algorithms, Reinforcement Learning, which is one of nature's most instinctive function and Transformers, which have revolutionized the age of NLP and sequence modelling into this particular sector are mere functions of mismatch of modules rather than lack of an entirely different architecture, no matter how much this report attempts to define this solution as one.

Existing attempts to relax the first assumption tend to reach for full Mixture-of-Experts routing across entire trading policies [4, 5, 6], which is parameter-heavy and occupies the routing problem at the wrong granularity: the whole policy, rather than the specific temporal and credit-assignment needs of a given asset at a given moment. Existing attempts to relax the second assumption [1, 2] bring in encoder-style pretraining and cross-asset attention, but for different purposes: EarnMore's masking is aimed at letting an investor swap assets in and out of the pool without retraining, and STHformer is a forecasting model rather than a decision-making one. Neither combines market-wide cross-asset representation with per-asset adaptive temporal routing inside a return-conditioned Decision Transformer. That combination is the space this proposal occupies.

2 Related Work and the Gap Being Filled

Table 1 summarises the closest prior work, what mechanism each uses, and specifically what it does not do. This is where the proposal's contribution sits.

Prior Work	Core Mechanism	What It Does Not Do
EarnMore [1]	Masked-autoencoder encoder/decoder + RL; masking supports customizable, dynamic stock pools.	Masking targets pool flexibility, not multi-scale temporal or credit-horizon representation; not a return-conditioned Decision Transformer.
STHformer [2]	Temporal encoder with cross-attention for stock relationships; pretrains via masked-sequence reconstruction.	A forecasting model, not an RL/sequence-decision policy; no notion of adaptive per-asset memory or credit assignment.
Action Q-Transformer [3]	Encoder for state value, decoder with action queries, used for interpretability in DRL.	Encoder/decoder split is generic and not finance-specific; no cross-asset attention or temporal-horizon routing.
MoEDRLPM / LLMoE / DeepUnifiedMom [4, 5, 6]	Mixture-of-Experts routing at the level of full trading policies or sub-models.	Routing operates on whole policies rather than temporal/credit-horizon needs of individual assets; heavier parameter footprint.
Vanilla Decision Transformer / PPO-FinRL [7, 14]	Return-conditioned sequence modelling or standard RL baselines for trading.	Treats assets independently or with a fixed context window; no market-wide representation or per-asset adaptivity.

Table 1: Closest prior work and the specific gap this proposal targets.

Read together, three gaps recur:

1. No prior system pairs market-wide, cross-asset representation learning with a return-conditioned Decision Transformer decision head.
2. No prior system adapts memory length and credit-assignment horizon per asset, per timestep, as opposed to per fixed policy or per fixed pool.
3. Existing Mixture-of-Experts trading systems route at the level of entire models, which is both more expensive and a coarser granularity than routing the temporal behaviour of individual assets.

3 Proposed Architecture

The pipeline has six stages. Market data is first compressed into a single market-wide representation, relationships between assets are made explicit through attention, each asset's current temporal/credit-horizon regime is identified, specialist Decision Transformers process each regime, and the specialists' outputs are fused into one portfolio decision.

3.1 Global Market Encoder

Inputs: OHLCV data, technical indicators, macroeconomic variables, current portfolio state, and sector membership.

The encoder is non-causal. Since it is not itself predicting the future, it can see the entire input window at once, which makes it a stronger feature extractor than a causal architecture would be [8, 9]. Its job is restricted to understanding the market as a whole, rather than beginning to make decisions.

3.2 Cross-Asset Attention

Every asset attends to every other relevant asset. The resulting attention matrix is effectively a dynamic, learned market graph: it can surface groupings such as a semiconductor ecosystem, a banking sector, an energy sector, supply-chain dependencies, or a macro-sensitive cluster, without any of these being hand-specified [9]. The graph is a by-product of training, useful for interpretation, not an objective the model is directly optimised toward.

3.3 Adaptive Routing

Using the enhanced, cross-attended embeddings, a routing module estimates each asset's current temporal behaviour along two axes, memory length and credit-assignment horizon, producing four regimes: Long-Memory/Long-Credit (LL), Long-Memory/Short-Credit (LS), Short-Memory/Long-Credit (SL), and Short-Memory/Short-Credit (SS).

Critically, this classifies an asset's *current* market behaviour, not a fixed identity: an index ETF will typically sit in LL, a momentum name in SS, and a stock approaching an earnings print in LS even if it normally behaves otherwise.

Prototype-Guided Routing. Rather than assigning explicit temporal-credit labels, the routing network is guided by a set of learnable prototype vectors,

$$C = \{c_{LL}, c_{LS}, c_{SL}, c_{SS}\},$$

representing latent temporal behaviors. Given an asset embedding z_i, routing probabilities are computed as

$$p_k = \frac{\exp(\mathrm{sim}(z_i, c_k))}{\sum_j \exp(\mathrm{sim}(z_i, c_j))}.$$

The prototypes are not treated as ground-truth classes but as soft behavioral priors that encourage similar assets to occupy nearby regions of the latent space. Training is performed jointly with the Decision Transformer using

$$\mathcal{L} = \mathcal{L}_{DT} + \lambda \mathcal{L}_{proto},$$

where $\mathcal{L}_{proto}$ regularizes the embedding space while the downstream decision objective determines the optimal routing.

3.4 Specialized Decision Transformers

Each regime is served by a Decision Transformer configured with different attention spans, context lengths, and reward horizons, rather than a single architecture stretched to cover every case [7]. Decision Transformers are employed as the final decision module because they formulate reinforcement learning as a sequence modeling problem rather than relying on temporal-difference bootstrapping. By conditioning actions on the desired return-to-go, historical states, and previous actions, they model long-range temporal dependencies directly through self-attention, enabling more effective temporal credit assignment than the incremental Bellman backups used in conventional TD methods. This design also avoids the instability associated with bootstrapped value estimation, permits reasoning over long decision horizons without discount-induced short-sightedness, and leverages the scalability and stable optimization characteristics of modern transformer architectures. Consequently, the proposed architecture combines adaptive market representation learning with a return-conditioned sequence model capable of generating portfolio decisions while preserving long-term dependencies across market trajectories.

3.5 Portfolio Fusion

The specialists' outputs are combined into one portfolio-level representation, so the final decision reflects individual asset signals, cross-sector interaction, and portfolio diversification together, rather than each asset's allocation being decided in isolation.

3.6 Portfolio Decision

The fused representation produces discrete actions (buy/sell/hold) or continuous portfolio weights, optimised under the Decision Transformer sequence objective [7], with a path to extend to a full RL objective [11, 12].

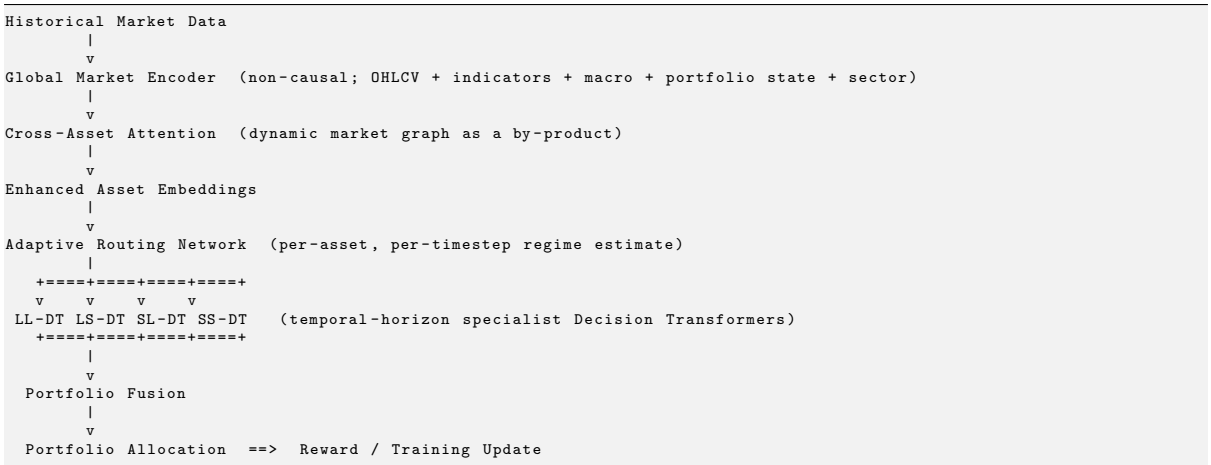


Figure 1: End-to-end pipeline: encode the whole market, make relationships explicit, route by current temporal regime, decide per specialist, fuse into one portfolio.

4 Design Note: A Change Worth Making Before Building This

One piece of the pipeline as sketched is worth revising, and it is worth being direct about why rather than building it as-is and discovering the problem during training.

Section 3.3–3.4 as described routes each asset to exactly one of four discrete expert Decision Transformers. This is a hard, effectively argmax-style routing decision. Two well-documented problems follow from that, both likely to bite here specifically:

1. **Discontinuous decisions:** Because the router re-classifies an asset's regime at every timestep, an asset sitting near a decision boundary can flip between experts from one step to the next. Each expert has its own context length and reward horizon, so a flip is not a smooth change in behaviour. It is a sudden change in what information the policy is even conditioning on. In a portfolio-allocation setting, that shows up as allocation instability that has nothing to do with the market actually changing.

2. **Training instability and cost:** Hard routing is not differentiable through the argmax, which historically pushes toward auxiliary load-balancing losses [13] and careful gating warm-ups to avoid expert collapse. It also means paying for four full Decision Transformers, which is a heavier parameter budget than the routing problem needs.

The fix preserves the core idea that different assets need different temporal and credit-assignment treatment without paying either cost. It makes the routing soft rather than hard. Concretely: replace the four discrete expert Decision Transformers with a small number of attention branches with different receptive fields (e.g. a short-range head and a long-range head) inside the encoder itself, and let a learned, per-asset gate blend their outputs continuously, rather than hard-assigning an asset to one branch. The decoder then still performs return-conditioned, causal decision generation as before, cross-attending into this single blended representation [7].

This keeps every one of the four original contributions (global encoding, cross-asset attention, adaptive temporal treatment, specialization) while removing the discontinuity and the four-times parameter cost. If discrete specialists are kept for interpretability reasons, at minimum a soft (weighted top-k) gate with a load-balancing auxiliary loss [13] and a temperature-annealing schedule should replace the hard argmax, so that regime boundaries are learned as soft transitions rather than cliffs.

5 Training Methodology

Training is split into two phases so that the expensive, data-hungry part of the problem (understanding markets) can use abundant, cheap data, while the part that genuinely requires trajectories stays lightweight.

1. **Phase 1: Encoder pretraining (cheap, abundant data).** Mask chunks of price, volume, and indicator history across the asset universe and train the Global Market Encoder plus Cross-Asset Attention to reconstruct them [8, 10]. No RL or trajectory data required at this stage; any historical market data suffices.

2. **Phase 2: Decision training (expensive, scarce data).** Train the Adaptive Routing module and the specialist Decision Transformers (optionally fine-tuning the encoder) on actual trajectory data under the Decision Transformer sequence loss [7], with a load-balancing term on the routing gate if soft routing is used as recommended in Section 4.

6 Research Gaps This Work Addresses

- Combines market-wide, cross-asset representation learning with a return-conditioned Decision Transformer decision head, a combination present in neither EarnMore [1] nor STHformer [2].

- Moves temporal/credit-horizon adaptivity from the level of a fixed model or a fixed asset pool down to the level of an individual asset's current behaviour, re-estimated per timestep.
- Places specialization inside the representation (multi-scale attention branches, per Section 4) rather than as full parallel policies, which is a lighter-weight and less-explored point in the design space than the whole-policy routing used by MoEDRLPM, LLMoE, and DeepUnifiedMom [4, 5, 6].
- Produces a dynamic, learned market graph as an interpretable by-product of training, without requiring a hand-specified sector taxonomy as ground truth.

7 Why This Should Outperform Existing Architectures

The case for this design over the alternatives rests on three points, each tied to a specific weakness in prior work rather than a general claim of novelty:

1. **Representation and decision are decoupled but still connected:** The encoder is free to be non-causal and therefore a better feature extractor [8], while the decoder stays causal and return-conditioned as a Decision Transformer must be [7]. Prior systems that use encoder/decoder splits (Action Q-Transformer [3]) do so for interpretability, not for this specific representation-quality argument, and finance-forecasting systems with strong encoders (STHformer [2]) are not decision-making policies at all.
2. **Adaptivity is cheaper and finer-grained than existing Mixture-of-Experts trading systems,** because (with the Section 4 revision) it lives inside attention branches of one encoder rather than across several full trading policies [13], and it is re-estimated per asset per timestep rather than fixed per policy.
3. **Cross-asset structure is learned rather than assumed:** Because the attention matrix is a by-product of training rather than a hand-built sector taxonomy, the model can represent relationships such as supply-chain links and macro-sensitivity clusters that a static sector label would miss [9], and it can do so before any decision is made, so every specialist already has sector-aware context.

The proposal does include a specific combination of various existing modules, and Section 2's table exists precisely so that claim can be checked against the nearest neighbours rather than asserted in the abstract.

8 Open Questions for Future Discussion

Kept brief and deliberately unresolved, these are the points still worth debating before or during implementation, not settled design choices:

- How many temporal scales belong inside the encoder, and should their receptive fields be fixed by design or learned end-to-end?
- Should decision training stay purely offline under the Decision Transformer sequence loss [7], or move to an Online Decision Transformer formulation for continued adaptation after deployment?
- What is the right ablation ladder to isolate each contribution's actual value? At minimum, it should compare a single-scale vs. multi-scale encoder, with vs. without encoder pretraining, and with vs. without cross-asset attention, removing each component one at a time against the same baselines.

9 Evaluation Plan

9.1 Baselines

- Vanilla Decision Transformer [7] (no market encoder, no cross-asset attention, no routing).
- EarnMore [1] (masked-autoencoder + RL, customizable pool).
- A standard PPO [11] or FinRL [14] baseline.

9.2 Ablations

- Single-scale vs. multi-scale attention branches in the encoder.
- With vs. without Phase 1 encoder pretraining (Section 5).
- With vs. without cross-asset attention (Section 3.2).
- Hard discrete routing vs. the soft-gated alternative proposed in Section 4, to directly test the claim that soft routing improves training stability and allocation smoothness.

9.3 Mathematical Formulation

Let the market contain N assets observed over the past T timesteps,

$$X = \{x_i^t\}_{i=1,t=1}^{N,T}, \qquad x_i^t \in \mathbb{R}^d.$$

The Global Market Encoder computes

$$E = f_{\text{enc}}(X),$$

where multi-head attention is

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V.$$

Cross-asset attention produces contextual embeddings

$$z_i = \sum_{j=1}^{N} \alpha_{ij} v_j, \qquad \alpha_{ij} = \frac{\exp(q_i^\top k_j/\sqrt{d})}{\sum_m \exp(q_i^\top k_m/\sqrt{d})},$$

forming a learned market relationship matrix $A = [\alpha_{ij}]$.
The routing network predicts the temporal-credit regime

$$r_i = g(z_i), \qquad r_i = [p_{LL}, p_{LS}, p_{SL}, p_{SS}],$$

with $\sum_k p_k = 1$. Under soft routing, specialist outputs are blended as

$$o_i = \sum_{k=1}^{4} p_k DT_k(z_i),$$

where each DT_k is a Decision Transformer specialized for one temporal-credit regime.
Portfolio representations are fused by

$$F = \sum_{i=1}^{N} w_i o_i, \qquad w_i = \frac{\exp(u_i)}{\sum_j \exp(u_j)},$$

and the policy predicts portfolio weights

$$\pi(X) = \{w_1, \ldots, w_N\}, \qquad \sum_i w_i = 1.$$

Training consists of encoder pretraining using masked reconstruction,

$$\mathcal{L}_{rec} = \frac{1}{|M|} \sum_{x \in M} \|x - \hat{x}\|_2^2,$$

followed by Decision Transformer optimization,

$$\mathcal{L} = -\sum_t \log P(a_t | R_{\leq t}, s_{\leq t}, a_{<t}) + \lambda \mathcal{L}_{route}.$$

The complete mapping is

$$X \to E \to Z \to DT \to F \to \pi(X).$$

10 Conclusion

This proposal makes use of many pre-existing ideas from the literature, including masked pretraining [8, 10], cross-asset attention [9], and expert specialization [13] all have precedent individually. The contribution is the specific combination of a market-wide, non-causal encoder with per-asset, per-timestep adaptive temporal routing feeding a return-conditioned Decision Transformer decoder [7]. Section 4's revision, moving from hard discrete expert routing to a soft, gated blend of multi-scale attention branches, is proposed as a change to the original sketch precisely because it keeps that combination intact while removing a specific, well-documented failure mode of Mixture-of-Experts systems [13]. The evaluation plan in Section 9 is designed so that every one of these claims, including the value of the encoder, cross-asset attention, and routing mechanism itself, can be checked individually rather than taken on faith.

The principal novel contribution of the proposal is the architectural design of the proposed algorithm, each of its module has been tested in various trivial, non trivial, deterministic and continuous temporal environments, both in a vaccum of their own regime and in relation with each other. Many of the individual building blocks of the architecture have been extensively studied and validated across a wide range of complex learning environments. Decision Transformers have demonstrated robust long-horizon sequential decision-making capabilities in challenging continuous-control benchmarks such as the MuJoCo locomotion suite, where agents learn stable policies under delayed rewards and high-dimensional state spaces. Transformer encoders have consistently shown exceptional representation-learning ability through self-supervised pretraining, while cross-attention mechanisms have been successfully employed in multi-agent environments, such as predator-prey (cat-and-mouse) settings, where the encoder learns how the behaviour of one agent is influenced by the actions and presence of another. This ability to model dynamic inter-entity relationships naturally extends to financial markets, where the movement of one asset is often conditioned on the behaviour of related assets through sectoral, macroeconomic, or supply-chain dependencies. Similarly, adaptive routing and expert-specialization mechanisms have repeatedly demonstrated their ability to allocate computational capacity efficiently across heterogeneous tasks.

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